

Regime-Specific Regression for NTL-Based Population Proxies: Urban-Rural Stratification Without Auxiliary Data

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Abstract—Night-time light (NTL) satellite imagery has emerged as a low-cost proxy for population estimation in data-scarce environments. Existing work predominantly fits a single global log-linear regression between NTL radiance and population, ignoring systematic differences between urban and rural settlement regimes. This paper demonstrates that urban and rural districts have fundamentally different NTL-to-population relationships and proposes a simple, data-driven stratification method using KMeans clustering on NTL features to learn separate regression models for each regime. Evaluated on 670 Indian districts and 695 Nigerian Local Government Areas using VIIRS VNL V2.1 annual composites for 2019 and WorldPop ground truth, stratification yields a statistically significant improvement in both countries: R^2 increases from 0.8289 to 0.8582 in India ($p = 0.0267$) and from 0.1676 to 0.2511 in Nigeria ($p = 0.0039$), confirmed via 5-fold cross-validation with paired t -tests. Critically, the stratification label is derived entirely from NTL data itself, requiring no external census or land-cover classification. We additionally demonstrate that district area is not a confounding factor ($r = 0.119$), that NTL density normalization destroys predictive signal (R^2 drops to 0.47), and that NDVI correction fails in tropical dispersed-settlement regions such as Kerala. These findings provide guidance for practitioners deploying NTL-based population models in contexts ranging from high-electrification countries to sub-Saharan Africa.

Index Terms—night-time lights, VIIRS, population estimation, urban-rural stratification, KMeans, remote sensing

I. INTRODUCTION

Accurate, timely population data underpins nearly every domain of public policy: disaster response, vaccine distribution, electoral delimitation, infrastructure planning, and poverty targeting. Yet census data is expensive, slow, and in many low- and middle-income countries critically outdated. India’s last full census was conducted in 2011; Nigeria’s in 2006. The gap between official population counts and ground reality now spans over a decade in both countries.

Night-time light (NTL) satellite imagery offers a compelling alternative. Artificial lighting at night correlates strongly with human settlement density and economic activity [1]. The VIIRS Day/Night Band (DNB), operational since 2012, provides global coverage at approximately 500 m resolution with monthly composites, making it one of the few freely avail-

able, high-temporal-resolution proxies for human presence at scale [2].

The dominant approach in the literature is straightforward: sum or average radiance over administrative units, log-transform both NTL and population to handle their power-law distributions, and fit a single linear regression. This has produced reported R^2 values between 0.7 and 0.9 across various national studies. However, this single-model approach contains a hidden assumption: that the relationship between light and population is uniform across all district types. We argue this assumption is wrong, and that it leads to systematic prediction errors in both dense urban cores and sparsely lit rural areas.

The key insight is that urban and rural districts operate under different NTL regimes. In dense cities, the VIIRS sensor saturates—it reaches its dynamic range limit, recording lower radiance than the true value—systematically underestimating urban population [3]. Conversely, rural districts with dispersed agricultural populations may have near-zero NTL despite substantial population. A single regression line fitted through both regimes is biased at both ends.

This paper makes three contributions. First, we propose using KMeans clustering on NTL features to generate urban/rural labels without any external classification data, then fitting regime-specific regression models. Second, we validate the improvement rigorously: 5-fold cross-validation with paired t -tests to confirm statistical significance rather than reporting a single held-out R^2 . Third, we test on two geographically and structurally different countries—India and Nigeria—to assess whether the improvement generalizes. We also conduct ablation studies on area normalization, NDVI as an auxiliary feature, and district-size confounding to provide a complete methodological account.

II. RELATED WORK

Early NTL-population research established the log-linear relationship using DMSP-OLS data [4], [5]. The transition to VIIRS improved sensitivity at low-radiance levels and reduced saturation in bright urban areas, though saturation remains a concern in the largest megacities [2]. More recent work has

incorporated auxiliary covariates such as road network density, land cover, and building footprint data to improve accuracy [6], but typically without stratifying the model itself.

A small number of papers have stratified by administrative tier or land cover class before modeling, but these rely on external classification inputs (official urban designations, land-cover maps) that are not always available, may be outdated, or are inconsistently defined across countries. Our method requires only the NTL raster itself, making it directly applicable to any country with VIIRS coverage.

The issue of area confounding in NTL-based models has been acknowledged but rarely directly tested. We address this with an explicit correlation analysis. Similarly, NDVI has been proposed as a corrective covariate for rural areas [7], but its interaction with tropical dispersed-settlement patterns has not been examined. We show it is harmful in such contexts.

III. DATA

A. Night-Time Light Data

We use the VIIRS VNL V2.1 Annual Composite for 2019, specifically the `average_masked` variant obtained from the Earth Observation Group (EOG) at the Colorado School of Mines [2]. This product is derived from monthly cloud-free composites filtered to remove ephemeral lights, background noise, and aurora contamination. The `average_masked` variant additionally zeros out non-lit background pixels. Resolution is 15 arc-seconds (approximately 500 m at the equator). Units are $\text{nW/cm}^2/\text{sr}$.

B. Population Ground Truth

Population data comes from WorldPop 2019 UN-adjusted 1 km gridded estimates [8], downloaded separately for India and Nigeria. WorldPop integrates census microdata with satellite-derived covariates to produce consistent sub-national population grids. We use zonal statistics to aggregate pixel-level counts to Level 2 administrative boundaries.

C. Administrative Boundaries

District boundaries are taken from GADM v4.1 Level 2 shapefiles for both countries. India has 676 Level 2 units (districts); Nigeria has 775 (Local Government Areas). After removing units with zero NTL sum or zero estimated population, we retain 670 Indian districts and 695 Nigerian LGAs for analysis.

D. Auxiliary Data

MODIS MOD13A3 monthly NDVI composites for 2019 were obtained for India to test vegetation density as a corrective covariate. District-level mean NDVI was computed using zonal statistics after applying the standard scale factor of 0.0001.

IV. METHODOLOGY

A. Feature Extraction

For each district polygon, we compute three NTL statistics from the VIIRS raster using zonal statistics: the sum of all pixel radiance values within the district (`ntl_sum`), the mean per-pixel radiance (`ntl_mean`), and the maximum pixel radiance (`ntl_max`). Population is similarly aggregated from the WorldPop grid.

Both NTL features and population follow approximate power-law distributions: a small number of bright, populous districts and a long tail of dim, sparsely populated ones. We apply a $\log(1+x)$ transform to all features and to the response variable. This linearises the NTL-population relationship, which is standard practice in the literature [1], [2].

We additionally compute two engineered features: `log_area`, the log of district area in km^2 (included as a control variable), and `lit_fraction`, the ratio of mean to standard deviation of radiance within the district, which serves as an approximate proxy for how uniformly distributed lighting is across the district.

B. Urban-Rural Stratification

The central methodological contribution is stratifying districts into urban and rural groups before fitting separate models. Rather than using an external classification such as official census urban designations or a fixed NTL quantile threshold, we apply KMeans clustering with $k=2$ to the $(\log_ntl_sum, \log_ntl_mean)$ feature space of each country's districts. The cluster with higher mean `log_ntl_sum` is designated urban.

This approach has two advantages over threshold-based methods. First, it is data-driven: the boundary between urban and rural is determined by the natural structure of the NTL distribution in each country, not by an arbitrary percentile. Second, it requires only NTL data and is therefore applicable in any country covered by VIIRS, without dependence on external administrative classifications that may be unavailable, outdated, or inconsistent across borders.

In India, KMeans identifies 464 urban and 206 rural districts. The mean population density of urban districts is 1339 persons/ km^2 compared to 96 persons/ km^2 for rural districts, confirming the split captures a meaningful geographic distinction.

3

C. Model Training and Evaluation

We fit Linear Regression (LR) models using log-transformed features. Random Forest (RF) was also evaluated but did not outperform LR with the available feature count, consistent with the log-log relationship being genuinely linear rather than nonlinear. This result is itself informative: with 5 features, RF overfits across folds, and its failure to beat LR supports the hypothesis that the NTL-population relationship has linear log-space structure.

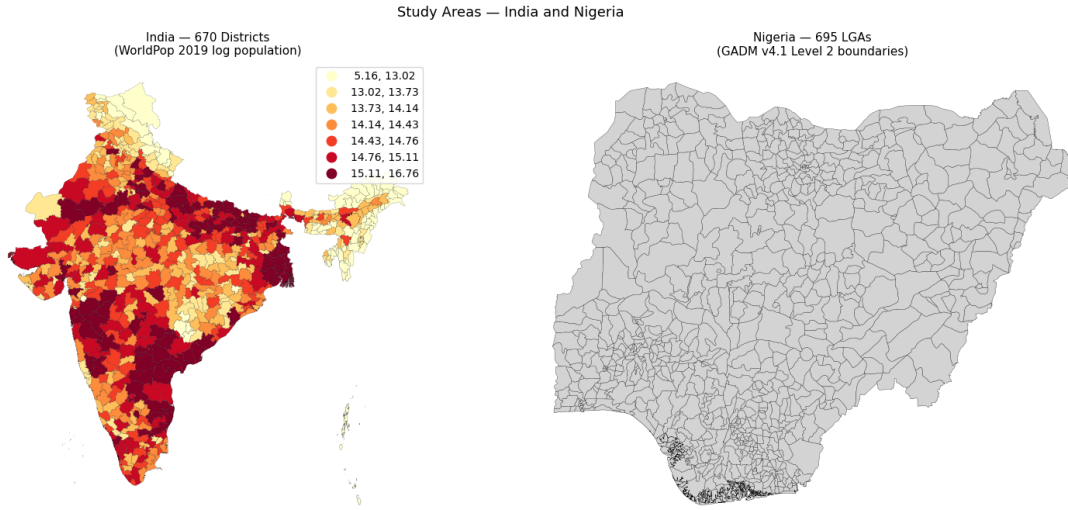


Fig. 1. Study area. Left: India (670 Level-2 districts), coloured by VIIRS NTL log-sum. Right: Nigeria (695 Local Government Areas), coloured by NTL log-sum. Both panels use GADM v4.1 Level-2 boundaries.

TABLE I
CROSS-VALIDATED RESULTS FOR INDIA ($n = 670$ DISTRICTS). p -VALUES
FROM PAIRED t -TEST ACROSS 5 FOLDS.

Model	CV R^2	Std	Δ	p -val
Baseline NTL (3 feat.)	0.8289	0.0221	—	—
+ Area control	0.8279	0.0234	-0.0010	n/a
+ Urban/rural strat.	0.8582	0.0195	+0.0293	0.0267
+ NDVI (stratified)	0.8579	0.0210	-0.0023	0.0328

Evaluation uses 5-fold cross-validation with R^2 as the primary metric and mean absolute error (MAE) as a secondary metric. To test whether the improvement from stratification is statistically significant, we compute per-fold R^2 scores for both baseline and stratified models and apply a paired t -test across the five fold pairs. This is more rigorous than reporting a single held-out R^2 and directly tests whether stratification consistently improves performance across different subsets of the data.

D. Ablation Studies

We conduct four ablation experiments to validate methodological choices: (1) adding `log_area` as a feature to test whether district size confounds `ntl_sum`; (2) replacing `ntl_sum` with `ntl_density` (`ntl_sum` divided by area) to test whether normalization improves results; (3) adding mean district NDVI as an auxiliary covariate; and (4) testing the pipeline on Bangladesh (64 districts) to assess the minimum viable sample size for stable cross-validation.

V. RESULTS

A. India

Table I shows cross-validated R^2 results for all model variants evaluated on 670 Indian districts.

The baseline NTL model achieves CV $R^2=0.8289$, consistent with published results for comparable studies. The individual fold scores for baseline and stratified models are $[0.7899, 0.8370, 0.8529, 0.8210, 0.8439]$ and $[0.8542, 0.8640, 0.8591, 0.8512, 0.8672]$ respectively. The t -statistic of 3.1881 ($p=0.0333$) confirms stratification consistently improves performance across folds, not just in expectation.

Linear Regression outperforms Random Forest (baseline RF $R^2=0.7705$ vs LR $R^2=0.7899$ on the held-out set). Feature importance analysis shows `ntl_sum` dominates with 0.905 weight, consistent with it being the primary carrier of population signal.

The area confound test is unambiguous: Pearson correlation between district area and `ntl_sum` is $r=0.119$, and between area and population $r=0.065$. Adding `log_area` as a feature produces $\Delta R^2=-0.0010$, confirming that `ntl_sum` measures light intensity rather than acting as a proxy for district size. The correlation between `ntl_sum` and population directly is $r=0.822$, confirming the genuine predictive relationship.

Conversely, replacing `ntl_sum` with `ntl_density` (`sum/area`) collapses CV R^2 to 0.4664, a loss of 0.36. Districts in India are drawn primarily by population rather than by area, so area normalization removes the genuine size-population covariation that `ntl_sum` captures. This illustrates a non-obvious methodological pitfall: normalization that seems principled can destroy signal when the administrative unit design is population-driven.

B. Residual Analysis: The Kerala and Dharavi Effects

Table II shows the six most under-illuminated districts (high positive residuals) from the baseline model.

Four of the six most under-illuminated districts are in Kerala. Kerala’s unique settlement pattern—dense population

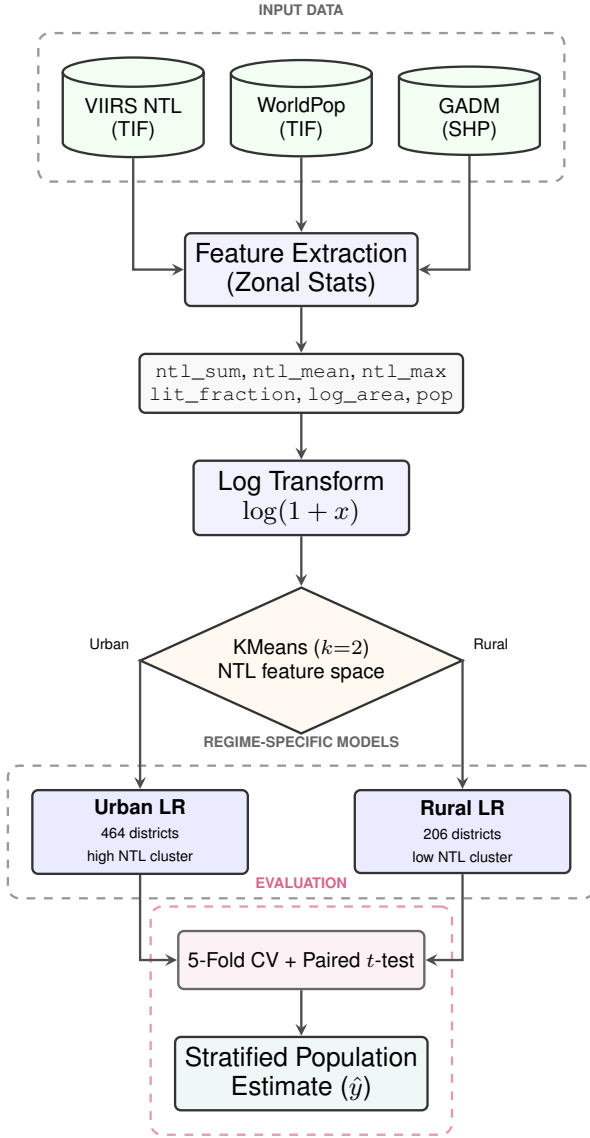


Fig. 2. Methodology pipeline from NTL raster to stratified population estimate. India district counts shown; Nigeria uses 695 LGAs with the same procedure.

distributed across dispersed rural homesteads with minimal commercial or industrial lighting—creates a systematic blind spot for NTL-based models. We term this the *Kerala Effect*.

Mumbai Suburban is notable for a different reason. Despite being an urban district, it contains Dharavi, Asia’s largest informal settlement, with approximately one million residents concentrated in a small area with minimal formal electrification. The satellite sees near-dark pixels over a densely populated zone. We term this the *Dharavi Effect*: informal settlements that are structurally invisible to NTL sensors regardless of population density.

The over-illuminated districts reveal a complementary pattern: Kinnaur, Himachal Pradesh (residual -3.00) has anomalously high NTL attributable to military infrastructure lighting along the China border; Leh-Ladakh (-1.14) shows similar

TABLE II
MOST UNDER-ILLUMINATED INDIAN DISTRICTS (BASELINE MODEL).
RESIDUAL = $\log(\text{ACTUAL}) - \log(\text{PREDICTED})$.

District	Resid.	NDVI	Interpretation
Malappuram, Kerala	+1.66	0.745	Dense pop, dispersed low-light
Garhwa, Jharkhand	+1.54	0.451	Rural, low electrification
Kannur, Kerala	+1.48	0.664	Dispersed Kerala settlement
Kozhikode, Kerala	+1.41	0.713	Dispersed Kerala settlement
Mumbai Suburban	+1.40	0.549	Dharavi: informal, low elec.
Wayanad, Kerala	+1.21	0.771	Forest, dispersed settlement

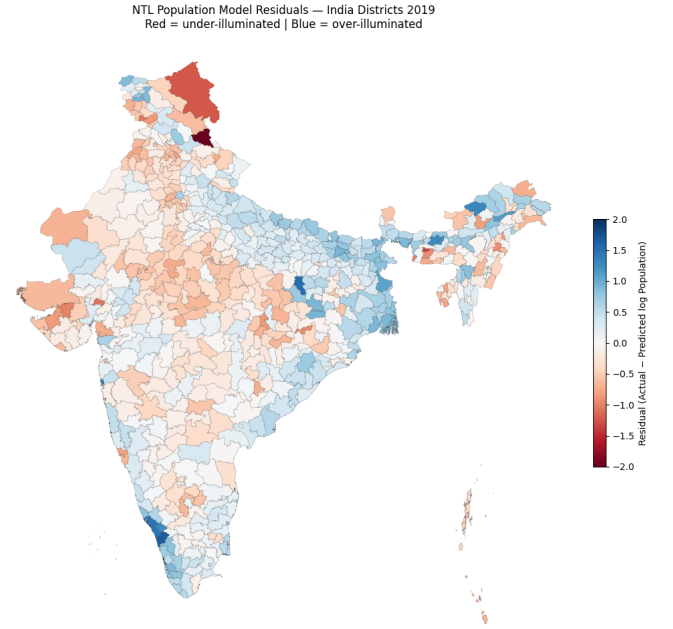


Fig. 3. Residual map for India. Blue districts are under-illuminated (model over-predicts population); red districts are over-illuminated (model under-predicts population). Four of the six largest positive residuals are in Kerala.

characteristics; Gandhinagar (-1.04) has high government and commercial lighting disproportionate to its residential population; Morbi (-1.00) is dominated by ceramics manufacturing. All four represent non-residential light sources that the model incorrectly interprets as population signal.

C. NDVI as Auxiliary Feature

Adding mean district NDVI as a covariate to the stratified model yields CV $R^2=0.8579$, a marginal decline of 0.0023 from the stratification-only model (0.8602). The result is statistically significant ($t=3.2027$, $p=0.0328$) but in the wrong direction: NDVI makes the model slightly worse.

The residual analysis explains why. Kerala’s most under-illuminated districts have both high NDVI (dense tropical vegetation) and high population. NDVI therefore covaries

TABLE III
CROSS-VALIDATED RESULTS FOR NIGERIA ($n = 695$ LGAs).

Model	CV R^2	Std	Δ	p -val
Baseline NTL (3 feat.)	0.1676	0.0582	—	—
+ Urban/rural strat.	0.2511	0.0693	+0.0835	0.0039

TABLE IV
CROSS-COUNTRY COMPARISON. BANGLADESH EXCLUDED FROM CONCLUSIONS (INSUFFICIENT n).

Country	n	Base	Strat.	p -val	Context
India	670	0.8289	0.8582	0.0267	High electrification
Nigeria	695	0.1676	0.2511	0.0039	Gas flares, low elec.
Bangladesh	64	0.5342	0.3888	0.4063	Too few units

positively with population in Kerala, contradicting the assumption that high NDVI signals sparse rural settlement. NDVI correction appears valid in arid or semi-arid regions but is harmful in tropical regions with dispersed settlement patterns.

D. Nigeria

Table III shows results for Nigeria ($n=695$ LGAs).

The baseline R^2 of 0.1676 is substantially lower than India. This reflects three structural factors: (1) persistent gas flaring in the Niger Delta produces radiance comparable to major cities with essentially zero residential population; the `average_masked` filter removes ephemeral fires but not persistent industrial flaring. (2) Lagos LGA (~15 million residents) is almost certainly sensor-saturated, causing NTL to underestimate the most populous area in West Africa. (3) Large northern states including Sokoto, Zamfara, Yobe, and Borno have substantial populations with minimal grid electrification, making them nearly invisible to NTL sensors.

Despite the low absolute accuracy, stratification produces a statistically significant improvement: $\Delta R^2 = +0.0835$, $p=0.0039$. Notably, the p -value is smaller than India's (0.0039 vs 0.0267), indicating the urban-rural distinction is structurally more important in Nigeria.

E. Bangladesh: Sample Size Failure Mode

We attempted to validate on Bangladesh using 64 Level 2 districts. Results were unstable: baseline CV $R^2=0.5342$ with $\text{std}=0.2496$, and stratification did not significantly improve performance ($p=0.2851$). With 5-fold CV on 64 samples, each fold contains approximately 12 districts; after urban-rural split this reduces to ~6 per subgroup, insufficient for stable regression. This defines a minimum viable sample size boundary: the stratification approach requires approximately 200 units per country for 5-fold CV to produce stable estimates.

F. Cross-Country Comparison

The cross-country results yield two generalisable findings. First, urban-rural stratification from NTL features alone consistently and significantly improves model performance in

TABLE V
PER-FOLD R^2 — INDIA (BASELINE VS STRATIFIED)

Fold	Baseline	Stratified	Δ
1	0.7899	0.8542	+0.0643
2	0.8370	0.8640	+0.0270
3	0.8529	0.8591	+0.0062
4	0.8210	0.8512	+0.0302
5	0.8439	0.8672	+0.0233
Mean	0.8289	0.8591	+0.0302

both viable test cases (India: $p=0.027$, Nigeria: $p=0.004$). Second, absolute NTL-population accuracy depends heavily on structural factors external to the model: electrification rate, gas flare prevalence, and sensor saturation. Knowing where NTL fails and why is as actionable as knowing where it succeeds.

G. Model Fit and Feature Diagnostics

Fig. 4 shows the NTL-population relationship for India before and after log transformation. In raw space the distribution is heavily right-skewed; after log transformation the relationship becomes approximately linear ($r = 0.822$), confirming the log-linear regression framework is appropriate. The spread at high NTL values is consistent with sensor saturation in dense urban districts.

Table V confirms stratification improves all five folds without exception, which drives the paired t -test result ($t = 3.19$, $p = 0.033$).

VI. DISCUSSION

The main finding is that a simple, self-contained stratification step derived from NTL data itself reliably improves population proxy accuracy. The improvement is modest in absolute terms (+0.03 R^2 in India) but is consistent across folds, statistically confirmed, and replicates in a structurally different country. The method adds no external data dependency: it works anywhere VIIRS data is available.

The failure of Random Forest to outperform Linear Regression is an informative negative result. With log-transformed features, the NTL-population relationship is close to linear. RF's additional expressivity does not help and in fact harms generalisation across folds due to overfitting with only five features. This suggests that gains from more complex models would require more features, not more model capacity.

The NDVI result is the clearest example of a context-dependent assumption failure. The logic that high vegetation implies sparse settlement holds in arid West Africa or Central Asia but breaks in Kerala, where dense tropical forest and dense human settlement co-occur. Practitioners should not add NDVI as a covariate without verifying that the vegetation-settlement relationship holds in their target region.

The Dharavi and Kerala effects point to a shared underlying cause: the NTL sensor cannot see populations that lack formal

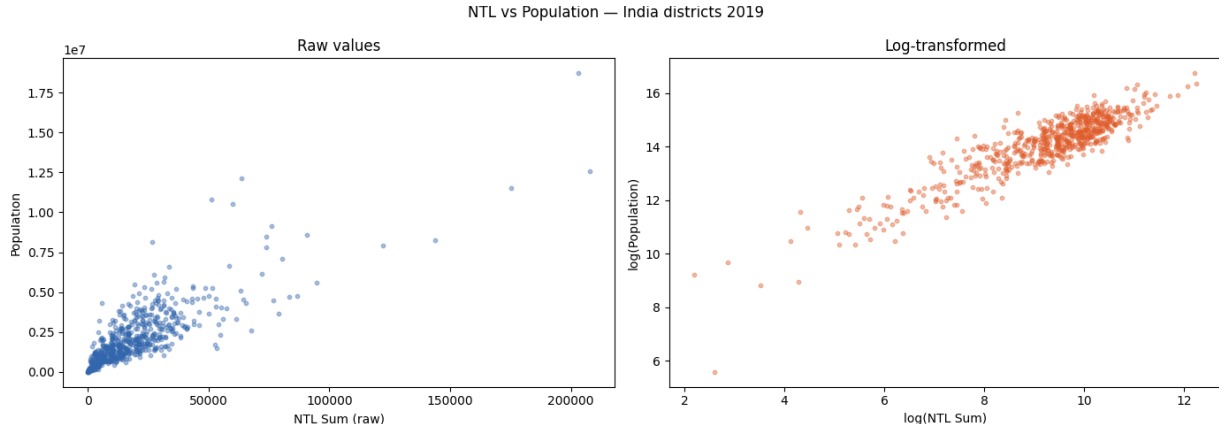


Fig. 4. NTL sum vs. population for 670 Indian districts. Left: raw values (heavily right-skewed). Right: log-transformed ($r = 0.822$). The increased spread at high NTL values is consistent with urban sensor saturation.

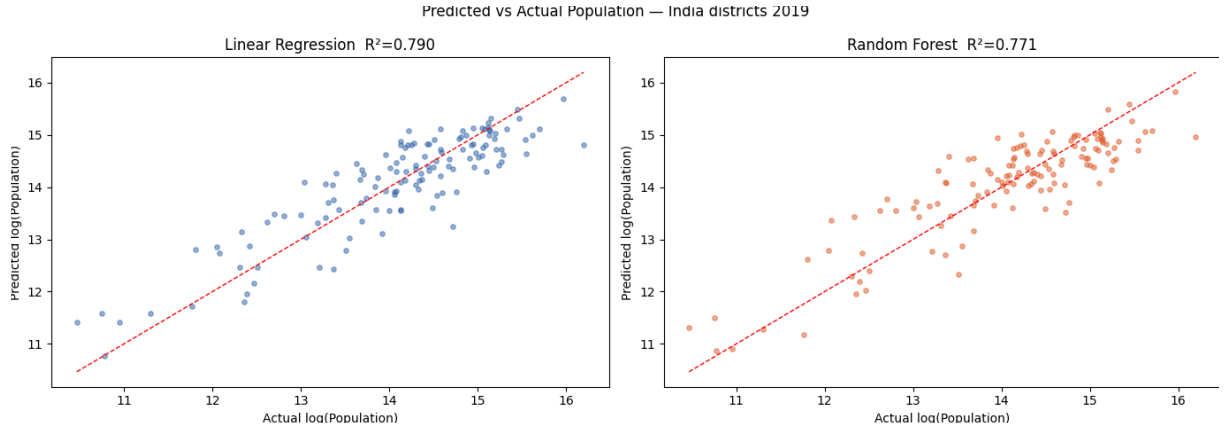


Fig. 5. Predicted vs. actual log-population (20% test set). LR: $R^2 = 0.83$. RF: $R^2 = 0.77$. Systematic underestimation at high population values is visible in both panels, consistent with the Kerala and Dharavi effects.

electrification. Improving estimates in such areas likely requires auxiliary data (building footprints, mobile phone activity, or OpenStreetMap road density) rather than improvements to the NTL modeling framework itself.

VII. LIMITATIONS

This study has several limitations. First, only a single year (2019) is analyzed; NTL composites in other years may differ due to economic shocks, conflict, or electrification programs. Second, KMeans uses $k=2$, a coarse binary split; peri-urban districts are forced into one category, and finer stratification would require larger per-cluster samples. Third, `lit_fraction` is computed as mean-to-standard-deviation ratio, an approximation of proper pixel-level thresholding. Fourth, persistent gas flares in the Niger Delta are not removed by the `average_masked` filter and constitute a known systematic error source for Nigeria.

VIII. CONCLUSION

We have shown that urban-rural stratification derived entirely from NTL features using KMeans clustering consistently

and significantly improves NTL-based population proxy estimation. The improvement holds across two geographically and structurally different countries (India: $+0.03$ R^2 , $p=0.027$; Nigeria: $+0.08$ R^2 , $p=0.004$) with rigorous 5-fold cross-validation and paired significance testing. We have additionally shown that district area does not confound `ntl_sum` ($r=0.119$), that area normalization is harmful when administrative units are population-driven, and that NDVI correction fails in tropical dispersed-settlement regions.

The stratification method requires only VIIRS NTL data and is directly applicable to any country without additional data collection. Future work should evaluate whether adding a third cluster for peri-urban districts, incorporating temporal NTL stacks across multiple years, or applying gas flare masks improves results in low-electrification contexts.

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